5 A stationary copper-64 nucleus undergoes a nuclear reaction to form a zinc-64 nucleus and a beta-particle. After the reaction, the zinc-64 nucleus moves off with negligible speed, while the beta-particle moves off with high speed.

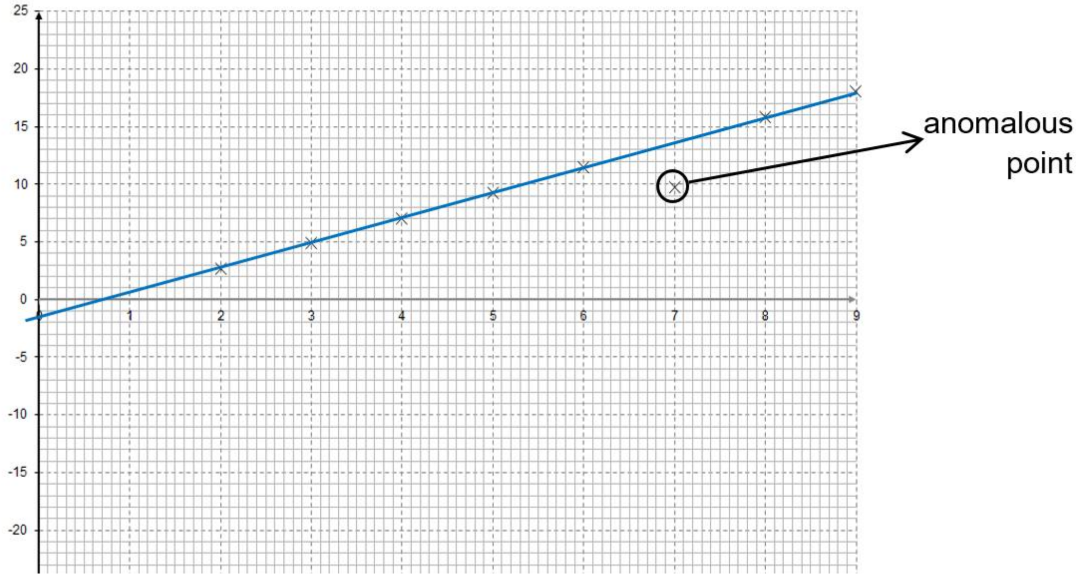
(a) Explain qualitatively why the beta-particle moves off with a much higher speed than the zinc-64 nucleus. [2]

By Conservation of Momentum, the **final total momentum of Zinc-64 nucleus and beta-particle must be equal to the initial momentum of Copper-64 which is zero**. [1] As the **beta-particle is much light** than Zinc-64 nucleus, the beta-particle will have much **larger speed** in order to have a momentum equal in **magnitude** but **opposite** in direction to that of Zinc-64 nucleus. [1]

(b) The kinetic energy E_K and the distance r of the beta-particle from the centre of nucleus of zinc-64 nucleus' centre are monitored. Fig. 5.2 shows how the values of E_K varies with $\frac{1}{r}$.

Assume that the beta-particle and zinc-64 nucleus form an isolated system.

For Examiner's Use



Identifying the anomalous point (circled and labelled) **and** sketching an appropriate BFL
Or
A BFL without anomalous point.

(i) By drawing an appropriate line on Fig. 5.2, determine the *y*-intercept, E_o, of the graph. [2]

Reading correctly the *y*-intercept: E_o = -2.0 x 10^{15} J

(ii) Explain the physical significance of the value of E_o found in **(b)(i)**. [2]

At $\frac{1}{r} = 0 \Rightarrow r \to \infty$. The vertical intercept represents the kinetic energy of the beta-particle when it is at an infinite distance from the Zinc-64 nucleus.

Since E_o is negative, it means that the beta-particle cannot escape from the Zinc nucleus' electric field / it requires additional k.e. to escape from the Zinc nucleus' electric field.

(iii) Assuming that total energy of the beta-particle remains constant during its motion, sketch labelled graphs on the axes in Fig. 5.2 to represent

1. the variation of the total energy E_T .

2. the variation of the electric potential energy E_P. [3]

1. Total Energy : E_T – a horizontal straight line passing through E_o.
2. EPE:

Treating the beta-particle and the Zinc-64 nucleus as an isolated system, the electric potential energy of the system $E_p = -\frac{1}{4\pi\varepsilon_0}\frac{Q_{Zn}q_\beta}{r} \propto -\frac{1}{r}$

When $r \to \infty \Rightarrow \dfrac{1}{r} = 0$, $E_p = 0$.

When $\dfrac{1}{r} = 9 \times 10^{13}\ \text{m}^{-1}$, $E_K = 18.0 \times 10^{-15}$ J,

$\Rightarrow$ Therefore $E_P = E_T - E_K = (-2-18) \times 10^{-15}$ J $= -20.0 \times 10^{-15}$ J

Hence, E_p graph is <u>a straight line graph with negative gradient</u>, passing through <u>(0,0) and (9, -20).</u>

[1]: a negative gradient straight line graph passing through (0,0)

[1]: The line passes through (9, -20.5) – note that shape of line must be correct first for the second mark to be awarded.

6 Fig. 6.1 shows a circuit consisting of two resistors connected in series to a d.c. supply.